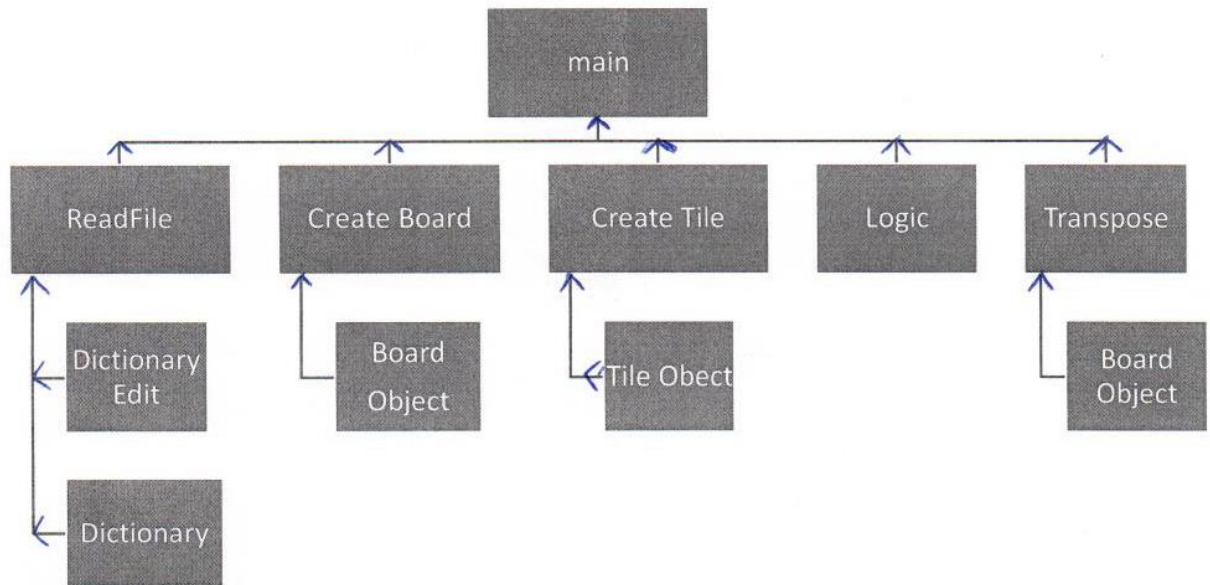


Solver:



1. This is the object diagram of the Class-objects used in the solver program.
2. The main class implements many objects such as
 - **Read File:** This class takes the dictionary file as input. The input file is later read with the while loop and the strings are added to dictionary created using the **Dictionary Edit** object. It has addWords function that adds words to the **Dictionary object**. **Dictionary Edit** is used to pass the strings to and it stores in Dictionary. **Dictionary** object is the blueprint for creating the dictionary used by DictionaryEdit. DictionaryEdit has other methods that search such as testing if a given string is valid or not.
 - **Create Board:** It reads the given board file and storing it in the array of **Board Object**. **Board Object** is the blueprint for data keeping each board tile must have such as storing a letter, word multiplier, letter multiplier, Boolean whether it has been played or not.
 - **Create Tile:** It reads the given file for tiles of each alphabet and used **Tile-Object** to store it. **Tile-object** is blue-print that stores information such as how many tiles, the letter, and the value of letter.
 - **Logic** class uses all the above given objects and uses them to do the back end move calculations and

bookkeeping. It first calculates the anchor points. Anchor points are points where it's possible to have next move. After calculating the anchor points it tests various strings via permutations and combinations to find the best possible string with highest score. After finding the highest score for horizontal move, it generates transpose to calculate the vertical move. It uses **Transpose object** to calculate the transpose. It takes a board and returns the transpose of a given board object and is called by main to generate transpose of the board.

3. Gui version also uses the same code structure except boardObject file was changed to meet the needs of given board. That lead to having the different GUILogic file because take board object input but there are two separate ones of those. GUILogic was later on altered to meet specific needs to GUI. Most of the GUI handling is done from main class inside the GUICode package.

